YASH DANIEL INGLE

Software Engineer (Backend & Platform) — Distributed systems, APIs, Python/Java, MySQL, Linux, CI/CD

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EDUCATION

Master of Embedded and Cyber-Physical Systems - University of California, Irvine, CA **GPA: 4.0 -** (December 2025)

(Systems Modelling, Control Systems, Wireless Sensor Networks, I/O Interface Programming, Embedded Linux/Yocto)

B. Tech - Computer Science with AI honors - MGM`s Jawaharlal Nehru Engineering College, India **CGPA: 9.16/10 -** (July 2024)

(Data Structures & Algorithms, OOPS, OS, DB, Computer Architecture, Computer Networks, Compilers, Distributed systems)

EXPERIENCE

Software Engineer at Atlas Copco | India ***Jan 2024 – June 2024***

(Python, Perl, Backend Systems, MySQL, RQ, CI/CD)

- Built **Python/Django** backend services for inventory/procurement workflows with clean **domain/service/data-access** boundaries; delivered **REST APIs** for teams with consistent contracts and **structured logging + metrics** for observability.
- Designed and tuned MySQL schemas/queries (indexes, pagination, EXPLAIN optimization) and built an RQ-based ingestion/sync pipeline with worker pools and staged validation→transform→load, processing **~200k jobs/day** and **~1M row updates/day** to enable reliable real-time SKU reporting.
- Integrated **legacy Perl ETL/ops scripts** for format normalization and batch exports, with retries and failure handling to keep real-time updates reliable while decoupling long-running jobs from API latency.
- Deployed Django services and RQ workers on **Linux servers via Docker**, and productionized delivery via **Git + Jenkins CI/CD** with unit/integration tests, a **regression test matrix**; automated **log/artifact capture**, reducing release cycle time by ~42%.

Co-Founder – Transfusiotrack (Blood Bag Delivery System) *(Full Stack, Flask, Docker, CI/CD)* [LINK] ***June 2023 – Present***

- Co-founded a healthcare logistics platform to reduce blood delivery delays; owned product + backend engineering, building **Flask REST services** with **MongoDB**; integrating **digital payments** for **100% cashless** operations across blood banks/hospitals.
- Architected the backend with clear **OOP boundaries** (domain/service/persistence) and optimized dispatch + tracking paths using **indexes, caching, pagination,** and **profiling**, improving delivery coordination by **~30%**.
- Built production-grade reliability for orders/payments using **idempotency keys**, **retry-safe workflows**, and **state-machine transitions** to ensure correctness under network failures & duplicate requests. Deployed and operated services with **Docker + Git-based CI/CD**, using **GitHub Copilot** to accelerate pipeline scripting & deployment hardening, cutting release cycles by ~30%.

Synaptics Incorporated - Apprentice | USA [LINK] ***March 2025- Dec 2025***

(Edge AI, Embedded & IOT, ARM + NPU, CV, Concurrency/threading, Image Computing Yocto Linux)

- Built a **wearable embedded navigation** *assistant for visually impaired users* on **Synaptics Astra SL1680**; led **Yocto** Linux bring-up (**headless deployment**) and developed on-device services in **C++/Python**, engineering a **3-stage concurrent pipeline** (capture → preprocess/infer → postprocess/control) with thread-safe queues + bounded buffers to achieve **<100 ms latency at ~1 W**.
- Integrated and stabilized the **Sony IMX477** camera stack at the **V4L2** level (nodes/params/validation); implemented robust long-running acquisition with **drop policies** and memory-safe buffering to prevent resource leaks and maintain throughput.
- Optimized compute and runtime reliability: applied **SIMD/NEON acceleration**, containerized inference/runtime with **Docker**, and added **structured logs + metrics** to improve observability and speed up field debugging and regression analysis.

Embedded System Intern at Envipco | USA *(Python Tooling, CV Integration)* ***Jul 2025 – Dec 2025***

- Built on-device computer vision for RVM *(Reverse Vending Machines)* using **OpenMV AE3** with **YOLOv8-style TFLite** inference and **NMS**; serialized detections (bottle/hand/logo + barcode ROI) over **UART/CAN** to drive the main controller **FSM**.
- Led **360° multi-camera barcode ring** evaluation (5 to 7 imagers), testing scanner/optics/firmware settings; built **Python tooling** (multi-serial orchestration + logging) to measure coverage gaps and **first-decode latency**, enabling sensor selection.

PROJECTS

Wafer Inspection Backend Simulator *(Python, Concurrency, APIs, Workflows)* ***Sept 2024 – Dec 2025***

- Built a **Python** backend workflow simulator for wafer inspection: ingested image tiles + metadata, executed a **multi-stage concurrent pipeline** using **thread-safe bounded queues/backpressure**, and exposed **APIs** for defect maps and run summaries.
- Implemented SCAN→PROCESS→CLASSIFY→REPORT **orchestration (FSM)** with **retries, fault recovery, idempotent state transitions**, and **structured logs/metrics** to support long-running runs and debugging.

AI-Powered Music Streaming App *(Flask/Firebase/Kotlin)* [LINK] **Jan 2023 – May 2023**

- Built an Android music streaming app backed by a **Flask REST API**, implementing **auth, playlists, and recommendation endpoints**; integrated **Firebase** for realtime updates and user/session data.
- Developed a **content-based recommender** using **TF-IDF feature vectors + KNN (cosine similarity)** to rank personalized tracks and playlist suggestions. Designed backend **data models + API contracts** for fast feed/playlist reads using **pagination** and **cache-friendly query patterns** to keep latency low.

SKILLS

Programming:	Python, C, C++, Java, Perl
Backend & Microservices:	REST APIs, Django, Flask, service design (OOP), API versioning
Distributed Systems & Concurrency:	multithreading, multiprocessing, worker pools, bounded queues/backpressure, idempotency
Data & Storage:	MySQL, MongoDB, schema design, indexing, query optimization, pagination
Systems & OS	Linux (Yocto, PREEMPT-RT), POSIX threads, RTOS profiling (gprof), debugging (gdb)
Tooling:	CoPilot, Git, Postman, CI/CD (GitLab/Jenkins), Docker, unit/regression (Unity/C, pytest)